

Aneurysmal Disease in Patients With Takayasu Arteritis

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ABSTRACT. **Objective.** Takayasu arteritis (TA) leads to stenotic disease. Aneurysmal lesions are rarer. This study assessed the main characteristics of aneurysmal disease in a Canadian cohort of patients with TA.

Methods. This monocentric retrospective study included patients with TA followed at the Mount Sinai Hospital Vasculitis Clinic in Toronto. Diagnosis of TA was based on clinical findings and/or satisfied the 1990 American College of Rheumatology classification criteria.

Results. Seventy-four patients were included. At any time, aneurysmal disease was found in 23 (31%) patients. Median disease duration was 9.0 (IQR 7.0-19.0) years. Prior hypertension ($P = 0.02$), fever ($P = 0.04$), and seizure disorders ($P = 0.03$) were more common. Limb claudication was less frequent ($P = 0.01$). Persistent and/or new aneurysms were demonstrated in 22/23 patients at follow-up. Thoracic aorta aneurysm (13/22) was most common, followed by abdominal aorta (8/22), subclavian (7/22), and carotid (6/22) artery disease. Aortic valve regurgitation was more frequent (9/23 vs 3/48; $P = 0.001$). Twenty-one patients had been treated with glucocorticoids (median 6.1 years [IQR 3.7-8.1]). Methotrexate, azathioprine, and leflunomide were repeatedly used. Infliximab (7/23) was used more often ($P = 0.04$), whereas tocilizumab was received by only 4 patients with aneurysmal disease ($P = 0.01$). Patients with aneurysms suffered more frequent relapses (2.0 [IQR 0.0-4.0] vs 1.0 [IQR 0.0-2.0], $P = 0.04$).

Conclusion. Aneurysmal disease was found in a significant proportion of patients with TA. Given that aneurysms may carry a risk of rupture, and are associated with a higher rate of relapse, this finding should be reported systematically in TA studies.

Key Indexing Terms: aneurysm, biologics, DMARDs, large-vessel vasculitis, relapse, Takayasu arteritis

Takayasu arteritis (TA), a large-vessel vasculitis, involves predominantly the aorta and its major branches.¹ Stenotic lesions and occlusions affect more than 90% of patients, and are the most

common types of arterial lesions. Such structural changes result from intimal hyperplasia, as well as media and intima fibrosis, secondary to chronic arterial wall granulomatous inflammation. Destructive changes of the arterial wall from oxidative stress and degradation by matrix metalloproteinases may lead to muscle cell loss, medial weakening, and eventually aneurysmal formation.^{2,3}

Few studies have reported features of patients with TA with aneurysms. In a large retrospective study of 2840 patients, patients with TA had a significant increased risk of aortic aneurysm (odds ratio [OR] 40.76).⁴ However, epidemiological data vary markedly, likely related to small sample size, patient selection biases, and the imaging modalities used. Radiologically, the reported frequency of aneurysmal disease varies from 4%^{5,6} to 46%,^{7,8} whereas by autopsy, more than half of patients with TA have been shown to have some form of arterial dilation.⁹ Male sex,^{6,10,11} hypertension,^{6,12} dyslipidemia,¹³ smoking,⁶ and disease duration > 5 years¹³ were variably reported as possible risk factors for aneurysmal disease.

Disease presentation, pattern of arterial involvement, as well as prognosis may vary according to race and ethnicity. Indeed, in Asian populations, aneurysmal lesions were described more frequently from the aortic root to the aortic arch.¹⁴ Two Chinese retrospective studies have described aneurysmal disease in TA; aneurysms were most common in the ascending aorta (33%) in one,⁶ whereas the other showed the abdominal aorta (22%) to be most frequently involved.¹¹ Multiple aneurysms were found

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in 13% to 30% of patients.^{6,11} Fatal outcomes associated with aneurysm formation have been described, including aortic valve regurgitation,^{6,11} heart failure, and aneurysm rupture.¹⁵⁻¹⁷

This study aimed to evaluate the prevalence, demographics, clinical manifestations, angiographic presentation, treatments, and outcomes of patients with arterial aneurysmal disease in a Canadian cohort of patients with TA.

METHODS

Patient population. This monocentric retrospective study included all patients with a diagnosis of TA, followed at the Mount Sinai Hospital Vasculitis Clinic in Toronto, Canada, from its inception to January 1, 2021. The diagnosis of TA was based on clinical presentation, laboratory measures and imaging studies, and/or fulfillment of the 1990 American College of Rheumatology (ACR) classification criteria.¹⁸ Enrollment in this study also required a minimum of 2 years of follow-up. Arterial aneurysms/dilatations were identified by reviewing reports of all available imaging.

Data collection. Baseline demographics and disease-specific characteristics were extracted from the center's research database (Canadian Vasculitis Research Network [CanVasc]), as well as from medical charts. Symptoms attributable to vasculitis, acute ischemic events, other cardiovascular (CV) complications, and laboratory abnormalities at diagnosis and during follow-up were analyzed. Vascular territory involvement and type of arterial lesions on imaging were collected at presentation and during follow-up. Medical treatments and revascularization procedures were also recorded.

Study definitions. Aneurysm was defined as arterial dilation to > 50% of the normal vessel diameter.¹⁹ Patients were then divided into 2 groups according to the presence or absence of aneurysmal disease. Dissection was defined as separation of vessel wall layers.¹⁹

Disease activity was defined according to the National Institutes of Health (NIH) criteria as new onset or worsening of ≥ 2 of the following features: (1) systemic features (eg, fever, musculoskeletal symptoms), (2) increased inflammatory markers (erythrocyte sedimentation rate [ESR] and/or C-reactive protein [CRP]), (3) features of vascular ischemia and/or inflammation (eg, claudication, diminished/absent pulse, bruit, vascular pain, limb asymmetric blood pressure), and/or (4) typical angiographic features (ie, new vascular lesion diagnosed by imaging in a previously unaffected vessel).²⁰

Disease relapse was defined as (1) ≥ 1 features of disease activity such as recurrent or new TA symptoms, new or progressive vascular lesion on imaging, and/or rise of inflammatory markers; (2) following a period of disease inactivity; and (3) requiring an intensification of or change in treatment.^{21,22}

Clinical disease pattern at onset was defined as in several previous studies.^{21,23} Progressive pattern corresponded to patients in whom > 1 year had elapsed between the first TA symptom and diagnosis. Acute disease was defined as a disease duration of < 1 year between the first symptom and TA diagnosis.

Successful treatment discontinuation implied the absence of any immunosuppressive therapy (ie, glucocorticoids [GCs], nonbiologic disease-modifying agents, and/or biologics) at the last visit for at least 6 consecutive months and the absence of active disease.

Statistical analyses. Categorical variables are presented as proportions (percentages). Continuous variables are presented as mean values (SD) for normally distributed variables or medians (IQR) otherwise.

Patients were divided into 2 groups according to the presence or absence of aneurysmal disease. Comparison between the 2 groups was performed by chi-square test or Fisher exact test, as appropriate, for categorical variables. Subgroup analyses were adjusted using the Bonferroni method. Comparison of continuous variables used the *t* test or Mann-Whitney *U* test, as appropriate. Analyses were 2-sided, with *P* values < 0.05 considered

significant. Statistics were carried out with SPSS software (version 27.0; IBM Corp.).

FL and CP had full access to all study data and take responsibility for its integrity and data analysis.

Ethics. Informed consent for participation in the CanVasc database was obtained from all patients (research ethics board [REB] #14-0052-D). This study was also specifically approved by the Mount Sinai Hospital (REB #21-0070-C).

RESULTS

Study population. One hundred six patients with a clinical diagnosis of TA were assessed at the Vasculitis Clinic during the study period. Thirty-two patients were excluded because of missing data, uncertain diagnosis, and/or they did not fulfill inclusion criteria. Seventy-four patients met study eligibility. Of these, 58% (43/74) satisfied the 1990 ACR TA classification criteria.¹⁸

Demographics and disease characteristics. Most patients were female (93%, 69/74) and White (42%, 29/69) or Asian (39%, 27/69). Median disease duration was 11.5 (IQR 7.0-19.0) years. Demographic features are further presented in Table 1.

Most had CV risk factors (66%, 49/74), including hypertension (49%, 36/74), dyslipidemia (31%, 23/74), obesity (15%, 8/55), and diabetes (11%, 8/74). Past tobacco use was recorded in 15% (11/73).

Clinical features, ischemic events, and vascular complications are presented in Table 2. More than half (64%, 47/74) of patients presented systemic complaints, with 1/6 patients reporting low-grade fever. Vascular complaints were described in 84% (62/74), mostly driven by limb claudication (72%, 53/74). Aortic complications included aneurysmal disease (23%, 17/74), valve regurgitation (17%, 12/71), and vessel dissection (5%, 4/74).

Vessel involvement and types of arterial lesions are presented in Table 3. Large-vessel involvement was demonstrated in most, with thoracic aorta (78%, 58/74), subclavian (86%, 63/73), and carotid disease (74%, 54/73) being most prevalent. Abdominal aorta, axillary, vertebral, and/or renal disease were found in less than half.

The majority (88%, 64/73) of patients were treated, at some point, with at least GCs (median duration 5.0 [IQR 2.9-9.2] years). Nine patients originally presented without active disease, and thus never received any treatment. Including the latter, 34 patients were off medication at their last visit after a median of 8.5 (IQR 3.7-16.5) years from diagnosis. Conventional immunosuppressants were given at some point in three-quarters of patients, including methotrexate (65%, 48/74) and azathioprine (49%, 36/74). A quarter of patients received biologics (23%, 17/74), mostly infliximab (IFX; 16%, 12/74). Tocilizumab (TCZ) therapy was given in 4 patients, and cyclophosphamide in 7 patients (Table 4).

The median number of vasculitis relapses was 1.0 (IQR 0.0-3.0) after a median disease duration of 11.5 (IQR 7.0-19.0) years. Revascularization was necessary in 30% (22/74) of patients. No aneurysmal ruptures or deaths were recorded (Table 1).

Comparative characteristics of patients with or without aneurysms. Aneurysmal disease was found in 31% (23) of the 74 study patients (Table 5). At diagnosis, 21% (15/73) of patients

Table 1. Demographic characteristics of 74 patients with TA.

	All Patients	Aneurysm	No Aneurysm	P
Gender				
Female	69/74 (93)	20/23 (87)	49/51 (96)	0.17
Race and ethnicity				0.02
White	29/69 (42)	9/21 (43)	20/48 (42)	
Indian	4/69 (6)	4/21 (19)*	0/48 (0)	
Other Asian	23/69 (33)	5/21 (24)	18/48 (38)	
Other	13/69 (19)	3/21 (14)	10/48 (21)	
Smoking status				
Nonsmoker	62/73 (85)	20/23 (87)	42/50 (84)	> 0.99
Ex-smoker	11/73 (15)	3/23 (13)	8/50 (16)	
CV risk factors				
Any risk factor	49/74 (66)	18/23 (78)	31/51 (61)	0.14
Diabetes mellitus	8/74 (11)	2/23 (9)	6/51 (12)	> 0.99
Dyslipidemia	23/74 (31)	5/23 (22)	18/51 (35)	0.24
Hypertension	36/74 (49)	16/23 (70)	20/51 (39)	0.02
Obesity	8/55 (15)	4/18 (22)	4/37 (11)	0.42
Medical history (prior to TA diagnosis)				
CAD	2/74 (3)	1/23 (4)	1/51 (2)	0.53
Heart failure	2/74 (3)	1/23 (4)	1/51 (2)	0.53
Cerebrovascular disease	3/74 (4)	2/23 (8)	1/51 (2)	0.23
Peripheral artery disease	2/74 (3)	1/23 (4)	1/51 (2)	0.53
Concomitant autoimmune disease				0.54
Any autoimmunity	16/74 (23)	4/23 (17)	12/51 (24)	0.76
IBD	5/16 (31)	2/4 (50)	3/12 (25)	
Seronegative arthritis	2/16 (13)	0/4 (0)	2/12 (17)	
Acute rheumatic fever	2/16 (13)	0/4 (0)	2/12 (17)	
Reactive arthritis	2/16 (13)	1/4 (25)	1/12 (8)	
Age at diagnosis, yrs, mean (SD); n patients	31.7 (11.8); 74	27.9 (10.9); 23	33.3 (11.9); 51	0.07
Disease duration, yrs, median (IQR); n patients	11.5 (7.0-19.0); 74	9.0 (7.0-19.0); 23	12.0 (7.0-19.0); 51	0.67
Disease relapse, median (IQR); n patients				
Any definition	1.0 (0.0-3.0); 73	2.0 (0.0-4.0); 23	1.0 (0.0-2.0); 50	0.04
New symptom	0.0 (0.0-2.0); 73	1.0 (0.0-3.0); 23	0.0 (0.0-2.0); 50	0.13
New sign	0.0 (0.0-0.0); 73	0.0 (0.0-0.0); 23	0.0 (0.0-0.0); 50	0.92
Changes on imaging	0.0 (0.0-1.0); 73	1.0 (0.0-2.0); 23	0.0 (0.0-1.0); 50	0.21
Rise in CRP/ESR	1.0 (0.0-2.0); 73	1.0 (0.0-3.0); 23	0.5 (0.0-1.0); 50	0.05
Clinical course				
Acute disease	27/72 (38)	10/22 (46)	17/50 (34)	0.36
Revascularization				
Any	22/74 (30)	10/23 (44)	12/51 (24)	0.08
Endovascular	13/74 (18)	3/23 (13)	10/51 (20)	0.74
Surgery	13/74 (18)	8/23 (35)	5/51 (10)	0.02
Type of procedure				
Aortic valve repair	3/74 (4)	2/23 (9)	1/51 (2)	0.23
Aorta repair	7/74 (10)	5/23 (22)	2/51 (4)	0.03
Heart revascularization	6/74 (8)	4/23 (17)	2/51 (4)	0.07
Renal PCI	2/74 (3)	0/23 (0)	2/51 (4)	0.22
Renal aneurysm repair	2/74 (3)	2/23 (9)	0/51 (0)	0.03
Other revascularization	12/74 (16)	2/23 (9)	10/51 (20)	0.32

Values are n/N (%) unless otherwise specified. Values in bold are statistically significant. * $P < 0.05$ for Indian Asian patients, as opposed to other subgroups, after adjustment using the Bonferroni method. CAD: coronary artery disease; CRP: C-reactive protein; CV: cardiovascular; ESR: erythrocyte sedimentation rate; IBD: irritable bowel disease; PCI: percutaneous coronary intervention; TA: Takayasu arteritis.

presented aneurysms, whereas at follow-up, this increased to 31% (22/72). Renal aneurysm had regressed in 1 patient under medical therapy alone. At the last follow-up, thoracic aorta aneurysm (18%, 13/71) was most prevalent, followed by abdominal aorta (11%, 8/71), subclavian (10%, 7/71), and carotid (9%, 6/71) aneurysmal disease.

Aorta vasculitis was described at some point in all 23 patients later diagnosed with arterial aneurysms. Thoracic aorta disease (including vessel thickening, stenosis, and/or occlusion) was most often reported, with aortic arch (87% [20/23] vs 45% [23/51], $P = 0.001$) and ascending aorta (61% [14/23] vs 35% [18/51], $P = 0.04$) being most prevalent (Table 3).

Table 2. Clinical presentation of 74 patients with TA.

	All Patients	Aneurysm	No Aneurysm	P
Symptoms				
Systemic symptoms	47/74 (64)	14/23 (61)	33/51 (65)	0.75
Weight loss	13/74 (18)	5/23 (22)	8/51 (16)	0.53
Low-grade fever	12/74 (16)	7/23 (30)	5/51 (10)	0.04
Fatigue	39/74 (53)	12/23 (52)	27/51 (53)	0.95
Myalgias	9/74 (12)	2/23 (9)	7/51 (14)	0.71
Arthralgias	14/74 (19)	2/23 (9)	12/51 (24)	0.20
Vascular symptoms	62/74 (84)	16/23 (70)	46/51 (90)	0.04
Headaches	25/74 (34)	9/23 (39)	16/51 (31)	0.51
Seizures	5/74 (7)	4/23 (17)	1/51 (2)	0.03
Visual changes	12/74 (16)	2/23 (9)	10/51 (20)	0.32
Carotidynia	15/74 (20)	7/23 (30)	8/51 (16)	0.21
Limb claudication	53/74 (72)	12/23 (52)	41/51 (80)	0.01
Erythema nodosum	2/74 (3)	1/23 (4)	1/51 (2)	0.53
Raynaud phenomenon	6/74 (8)	1/23 (4)	5/51 (10)	0.66
Ischemic complications				
Any	22/74 (30)	5/23 (22)	17/51 (33)	0.31
Cerebrovascular accident	14/74 (19)	3/23 (13)	11/51 (22)	0.53
Ischemic limb (requiring revascularization)	4/74 (5)	0/23 (0)	4/51 (8)	0.30
Blindness	1/73 (1)	0/23 (0)	1/50 (2)	> 0.99
CAD/myocardial infarction	9/74 (12)	4/23 (17)	5/51 (10)	0.45
Mesenteric ischemia	2/74 (3)	0/23 (0)	2/51 (4)	> 0.99
Other complications				
Renovascular hypertension	24/74 (32)	8/23 (35)	16/51 (31)	0.77
Aortic complications (any)	20/74 (27)	15/23 (65)	5/51 (10)	< 0.001
Aortic regurgitation	12/71 (17)	9/23 (39)	3/48 (6)	0.001
Aortic dissection	4/74 (5)	1/23 (4)	3/51 (6)	> 0.99
Aortic aneurysm	17/74 (23)	17/23 (74)	0/51 (0)	< 0.001
Heart failure	1/74 (1)	1/23 (4)	0/51 (0)	0.31
Pulmonary hypertension	1/71 (1)	0/23 (0)	1/48 (2)	> 0.99
Laboratory				
Increased inflammatory markers	27/62 (44)	6/20 (30)	21/42 (50)	0.14
Elevated CRP	16/55 (29)	5/19 (26)	11/36 (31)	0.74
Elevated ESR	23/61 (38)	6/20 (30)	17/41 (42)	0.39
Laboratory measures (at diagnosis), median (IQR); n patients				
CRP, mg/L	21.5 (6.3-54.3); 44	41.0 (17.2-67.0); 15	12.0 (5.6-47.0); 29	0.18
ESR, mm/h	40.0 (15.5-76.5); 49	55.0 (42.0-103.0); 14	33.0 (14.0-60.0); 35	0.04
Laboratory measures (after 2 yrs), median (IQR); n patients				
CRP, mg/L	4.0 (1.1-8.0); 43	2.2 (0.9-10.2); 18	4.9 (1.6-6.9); 25	0.63
ESR, mm/h	18.5 (9.3-27.5); 40	20.0 (5.0-30.0); 13	18.0 (10.0-25.0); 27	0.95

Values are n/N (%) unless otherwise specified. Values in bold are statistically significant. CAD: coronary artery disease; CRP: C-reactive protein; ESR: erythrocyte sedimentation rate; TA: Takayasu arteritis.

Significant differences ($P = 0.02$) in race and ethnicity could be demonstrated between patients with and without aneurysm. More patients from India (19% [4/21] vs 0% [0/48], $P < 0.05$) had arterial aneurysms (Table 1).

Patients with aneurysms demonstrated numerically higher CV risk factors. This was mostly driven by increased number of patients with prior hypertension (70% [16/23] vs 39% [20/51], $P = 0.02$; Table 1).

Fever, at any timepoint, was more frequent in patients with TA with aneurysms (30% vs 10%, $P = 0.04$) who also demonstrated fewer vascular symptoms (70% vs 90%, $P = 0.04$), such as limb claudication (52% vs 80%, $P = 0.01$). Conversely, seizure disorder was more frequent in the aneurysmal group (17% vs 2%, $P = 0.03$; Table 2).

ESR at diagnosis of TA was higher in patients with aneurysms ($P = 0.04$). Differences in CRP were nonsignificant, as were laboratory values after 2 years of follow-up (Table 2).

At the last visit, most patients (65%) with aneurysmal disease were still requiring immunosuppressive therapy. Tumor necrosis factor inhibitors (TNFi), mostly IFX, were more often prescribed in patients with aneurysmal disease (35% [8/23] vs 12% [6/51], $P = 0.03$). TCZ was received only by patients with aneurysmal disease (17% [4/23], $P = 0.01$). No significant differences were found with other immunosuppressants (Table 4).

Patients with aneurysms suffered more vasculitis relapses (2.0 [IQR 0.0-4.0] vs 1.0 [IQR 0.0-2.0], $P = 0.04$; Table 1). Aortic complications were more frequent in the aneurysmal group, driven by increased rates of aortic aneurysms (74% [17/23] vs

Table 3. Vessel involvement in 74 patients with TA.

	All Patients	Aneurysm	No Aneurysm	P
Vessel involvement				
Ascending aorta	32/74 (43)	14/23 (61)	18/51 (35)	0.04
Aortic arch	43/74 (58)	20/23 (87)	23/51 (45)	0.001
Descending aorta	44/74 (60)	16/23 (70)	28/51 (55)	0.23
Thoracic aorta (any involvement)	58/74 (78)	22/23 (96)	36/51 (71)	0.02
Suprarenal abdominal aorta	28/73 (38)	10/22 (46)	18/51 (35)	0.41
Infrarenal abdominal aorta	31/72 (43)	9/21 (43)	22/51 (43)	0.98
Aorta (any involvement)	65/74 (88)	23/23 (100)	42/51 (82)	0.05
Carotid artery	54/73 (74)	16/22 (73)	38/51 (75)	0.87
Vertebral artery	29/69 (42)	6/21 (29)	23/48 (48)	0.13
Subclavian artery	63/73 (86)	18/22 (82)	45/51 (88)	0.48
Axillary artery	20/41 (49)	4/9 (44)	16/32 (50)	> 0.99
Celiac artery	25/73 (34)	9/22 (41)	16/51 (31)	0.43
Mesenteric artery	24/73 (33)	6/22 (27)	18/51 (35)	0.50
Renal artery	32/74 (43)	9/23 (39)	23/51 (45)	0.63
Iliac artery	12/45 (27)	2/12 (17)	10/33 (30)	0.47
Femoral artery	5/20 (25)	1/5 (20)	4/15 (27)	> 0.99
Coronary artery	8/27 (30)	4/13 (31)	4/14 (29)	> 0.99
Pulmonary artery	9/59 (15)	5/21 (24)	4/38 (11)	0.26
Type of lesion				
Thickening	61/74 (82)	21/23 (91)	40/51 (78)	0.32
Stenosis	70/74 (95)	21/23 (91)	49/51 (96)	0.58
Occlusion	47/74 (64)	12/23 (52)	35/51 (69)	0.17

Values are n/N (%). Values in bold are statistically significant. TA: Takayasu arteritis.

Table 4. Treatment of 74 patients with TA.

	All Patients	Aneurysm	No Aneurysm	P
Treatment				
Free of Rx	34/74 (46)	8/23 (35)	26/51 (51)	0.20
Duration off Rx, yrs, median (IQR); n patients	8.5 (3.7-16.5); 29	10.8 (5.5-17.0); 7	7.1 (3.5-16.3); 22	0.60
Steroid use				
Pulse methylprednisolone	7/67 (10)	4/21 (19)	3/46 (7)	0.19
Oral glucocorticoids	64/73 (88)	21/23 (91)	43/50 (86)	0.71
Duration, yrs, median (IQR); n patients	5.0 (2.9-9.2); 62	6.1 (3.7-8.1); 21	4.3 (2.1-9.2); 41	0.33
Initial prednisone dose				
> 40 mg/d	44/58 (76)	14/18 (78)	30/40 (75)	0.82
< 40 mg/d	14/58 (24)	4/18 (22)	10/40 (25)	> 0.99
Other treatments				
cDMARDs	56/74 (76)	17/23 (74)	39/51 (77)	0.81
Methotrexate	48/74 (65)	16/23 (70)	32/51 (63)	0.57
Azathioprine	36/74 (49)	11/23 (48)	25/51 (49)	0.92
Leflunomide	15/74 (20)	7/23 (30)	8/51 (16)	0.21
Mycophenolate mofetil	4/74 (5)	2/23 (9)	2/51 (4)	0.58
Cyclophosphamide	7/74 (10)	4/23 (17)	3/51 (6)	0.19
Any biologic	17/74 (23)	8/23 (35)	9/51 (18)	0.11
TNFi (any)	14/74 (19)	8/23 (35)	6/51 (12)	0.03
Infliximab	12/74 (16)	7/23 (30)	5/51 (10)	0.04
Adalimumab	1/74 (1)	1/23 (4)	0/51 (0)	0.31
Etanercept	2/74 (3)	1/23 (4)	1/51 (2)	0.53
Tocilizumab	4/74 (5)	4/23 (17)	0/51 (0)	0.01
Ustekinumab	1/74 (1)	1/23 (4)	0/51 (0)	0.31
Abatacept	2/74 (3)	0/23 (0)	2/51 (4)	> 0.99
Rituximab	2/74 (3)	1/23 (4)	1/51 (2)	0.53

Values are n/N (%). Values in bold are statistically significant. cDMARD: conventional disease-modifying antirheumatic drug; TA: Takayasu arteritis; TNFi: tumor necrosis factor inhibitor.

Table 5. Aneurysm anatomy in 74 patients with TA.

Vessel Involvement	At Diagnosis	At Follow-Up
Any	15/73 (21)	22/72 (31)
Aortic root	3/69 (4)	2/71 (3)
Ascending aorta	8/69 (11)	9/69 (13)
Aortic arch	7/69 (10)	5/69 (7)
Descending aorta	5/69 (7)	7/71 (10)
Thoracic aorta (any involvement)	13/70 (19)	13/71 (18)
Suprarenal abdominal aorta	3/66 (5)	7/71 (10)
Infrarenal abdominal aorta	2/60 (3)	5/70 (7)
Aorta (any involvement)	13/71 (18)	15/71 (21)
Carotid artery	5/64 (8)	6/71 (9)
Vertebral artery	0/60 (0)	1/66 (2)
Subclavian artery	2/67 (3)	7/71 (10)
Axillary artery	0/29 (0)	1/38 (3)
Celiac artery	2/64 (3)	3/70 (4)
Mesenteric artery	2/64 (3)	3/69 (4)
Renal artery	2/65 (3)	2/71 (3)
Iliac artery	0/27 (0)	1/40 (3)
Femoral artery	0/11 (0)	0/13 (0)
Coronary artery	0/12 (0)	0/23 (0)
Pulmonary artery	1/45 (2)	4/55 (7)

Values are n/N (%). TA: Takayasu arteritis.

0% [0/51], $P < 0.001$), as well as aortic valve regurgitation (39% [9/23] vs 6% [3/48], $P = 0.001$; Table 2). Aorta repair in patients with TA with aneurysmal disease was also more frequently performed (22% [5/23] vs 4% [2/51], $P = 0.03$; Table 1). No differences were demonstrated in ischemic complications, nor in rates of aortic dissection.

DISCUSSION

In this monocentric retrospective study of patients with TA with extended follow-up, aneurysmal disease was demonstrated in almost a third of patients. Prior hypertension, seizure disorders, low-grade fever, and aortic valve regurgitation were more common in patients with aneurysms. Aorta vasculitis involvement had been diagnosed previously in all patients with subsequent arterial aneurysmal lesion. Thoracic aortic aneurysm was most common, followed by abdominal aortic aneurysm, though subclavian, brachiocephalic, and carotid aneurysms were demonstrated. Most patients with aneurysms were treated with GCs, and a substantial proportion also required further immunosuppressives, including IFX. Despite this, patients with TA with aneurysms suffered more vasculitis relapses. To our knowledge, the latter has not been previously described.

Aneurysm prevalence in TA varies considerably. Approximately 31% of our cohort showed arterial aneurysms. Multiple factors may come into play. Race and ethnicity have been shown to affect clinical presentation, as well as angiographic involvement. Our patient population consisted primarily of White patients (43%), with less than a quarter (24%) being of Asian descent (other than India). Two Chinese monocentric retrospective studies reported prevalences of 4.2% to 16.6%.^{6,11} Our cohort included only 5 patients of Chinese or Korean descent, of which only 2 had arterial aneurysms. Interestingly, all 4 Indian patients had

aneurysmal disease ($P < 0.05$). Older reports suggested a predilection for thoracic aorta aneurysm in Indian patients with TA,²⁴ although more contemporary studies showed varying frequencies (9–22%).^{12,25} Similar to our findings, mixed race cohorts reported higher incidences (France [12–24%],^{21,26} South Africa [47%],⁸ and the US [27%]²⁰). Further, in our cohort, aneurysms were found in 10 pediatric patients, with a mean age at disease onset of 13.4 (SD 3.7) years. Aneurysmal lesions have previously been reported in 13% to 65% of pediatric patients with TA.^{11,27,28} In 1 study, the interval between pediatric TA onset and aneurysm detection was only 2 years.¹¹

Imaging modalities and duration of follow-up may also contribute to variable frequencies. Radiological^{7,29} and surgical^{16,30} TA cohorts, in which there is systematic screening at diagnosis and periodic imaging thereafter such as in our center, report aneurysm frequency of 32% to 45% and 48% to 64%, respectively. This is further supported by the high frequency of arterial aneurysm (up to half) recorded in autopsy series.^{2,31} The duration of follow-up in our study was considerable (median 11.5 [IQR 7.0–19.0] years), as opposed to other cohorts with a median of 36.0 to 72.0 months.^{6,11} Only 1 other small surgical retrospective study reported outcomes of patients with TA with operated thoracic aortic aneurysm, with a follow-up of 7.4 years.³²

Aneurysm development in TA is likely multifactorial, and possibly not solely due to vessel wall inflammation. In patients who are nonarteritic, aneurysms are most often degenerative and arise in association with traditional CV risk factors. High blood pressure was diagnosed in more than two-thirds of our patients with aneurysms, significantly more often than in patients without. Similar rates of hypertensive disease in patients with TA with aneurysms have previously been described.^{6,7,11,12,25,29,30} Interestingly, and as previously reported by others, renovascular hypertension was not more frequent in our patients who experienced aneurysms.^{5,7,11,12} Though 1 recent Chinese retrospective study demonstrated increased risk of aneurysm development in patients with TA with elevated low-density lipoprotein,¹³ our cohort, as well as others, did not demonstrate such an association.^{6,7,11} Yang et al hypothesized whether tobacco use, at least in men, was a possible contributor to aneurysmal disease, as 10 of 14 male patients with TA with aorta aneurysms were active smokers.⁶ In our cohort, as well as others,^{11,13} no difference was found in tobacco use between groups.

Large-vessel inflammation likely contributes to aneurysmal disease process. Similar to our findings, Pan et al reported that a third of their patients with TA with aneurysms presented fever, significantly more than in patients without (OR 6.79).¹³ Previous reports also suggested higher inflammatory markers in patients with TA with arterial aneurysm,^{6,11,25,32} with 1 other study demonstrating increased risk of aneurysm (OR 2.85) with ESR elevation.¹³

Previous reports have noted variable aneurysm location, with some demonstrating increased rates of thoracic aorta aneurysm (1–37%),^{6,8,13,26,29,33–36} whereas others showed the abdominal aorta (1–47%) to be more frequently involved.^{7,11,12,25,37} At the last follow-up, close to a quarter of our cohort was diagnosed with large-vessel aneurysmal disease. Thoracic aorta

aneurysms (18%) were most prevalent, notably the ascending aorta, followed by abdominal aorta aneurysms (11%). Cheng et al described a higher proportion of Numano Classification type IIa (ie, ascending aorta, aortic arch, and branches) vasculitic involvement in patients with TA with arterial aneurysms, which is similar to our findings.¹¹ Aortic valve regurgitation, a known poor prognostic risk factor,^{15,37-39} was also noted more frequently in our cohort with aneurysms, in accordance with previous reports (5-33%).^{11,21,25,40} No differences in total burden of arterial stenosis and/or occlusion, nor in peripheral vessel involvement, were found, barring a nonsignificant decrease of iliac artery involvement. However, clinical limb claudication was less frequent in our patients with aneurysmal disease. Such a finding has previously been reported.^{11,13} Subclavian and iliac artery involvement were present, respectively, in up to 10% and 3% of our cohort, and 1% to 15% and 0% to 12% in previous studies.^{6,35,37} Seizures were more frequently recorded in our cohort of patients with aneurysmal disease. Rates of carotidynia, visual complaints, headaches, and cerebrovascular accidents did not differ significantly between groups. Carotid and vertebral artery aneurysms were found in up to 9% and 2% of our patients, respectively. Rates of extracranial artery aneurysm in TA are quite variable, with 1 report describing them as extremely rare, whereas carotid arteries were described as the second most common site of aneurysm in another report.^{11,37,41,42}

Management of aneurysms in TA seldom used immunosuppressive therapy, other than systemic GCs. In previously presented studies,^{6,11} GCs were used in 62% to 86% of patients with TA with aneurysms. However, immunosuppressives were given in only 3% to 15%. Only Pan et al employed immunosuppressants in two-thirds of their cohort, mostly cyclophosphamide (43%), TCZ (29%), and methotrexate (24%).¹³ Of note, use of TNFi in patients with TA with aneurysms was not extensively reported. The majority of our patients with TA with aneurysms (91%) were treated, at some point, with at least GCs. Conventional immunosuppressants were given in close to three-quarters of these patients, whereas biologics were prescribed in more than a third, with IFX being most frequent. In our cohort, tocilizumab was given only to patients with aneurysms. Revascularization was required in less than half of patients, of which aorta repair was necessary in less than a quarter. No aneurysmal rupture, nor deaths, were recorded. It is not well known whether immunosuppression can play a role in preventing aneurysm growth. In Pan et al, after a median follow-up of 18.3 months on immunosuppressive therapy, 15% of aneurysms showed no change and up to 39% actually demonstrated reduction in size.¹³ Sueyoshi et al studied aortic aneurysm outcome in TA by imaging and reported aneurysm expansion in 53% of their cohort.⁷ After a median of 36.0 months, 3 out of 9 patients with TA with an increase of aneurysm size demonstrated rapid expansion and eventual rupture.⁷ In patients with TA with aneurysmal disease, frequency of rupture is estimated at 3% to 11%.^{8,29} The majority of patients with TA with aneurysm (80%) eventually require revascularization,¹¹ with aneurysm operations being performed in more than half (51%).⁶ Although a progressive increase in the number of aneurysms was noted in our study, considering

the lack of aneurysmal rupture with low rate of surgical aorta repair, one may wonder whether immunosuppressive therapy may indeed reduce, at least partially, aneurysm progression.

Despite comparable clinical course, vasculitis relapses were more frequent in our patients with aneurysms. Such a finding, to our knowledge, has not previously been published. Patients with TA with aneurysmal disease may indeed suffer from more severe vasculitis. In the latter group, we demonstrated more frequent fever, seizures, and higher ESR at diagnosis. Other cohorts have also reported such findings,^{6,11,13,25,32} as well as a higher prevalence of elevated Indian Takayasu Clinical Activity Score (ITAS2010).¹¹ Prescription of TNFi and interleukin 6 receptor antagonist could also be considered as a surrogate marker of severity, likely explaining increased use of biologics in patients with TA with aneurysms. Similar to previous reports,^{20,21,35} more than half of our patients with TA experienced relapse during their disease course, even though more than a third were able to taper off therapy for at least 6 months.⁴³ Successful treatment discontinuation did not seem to alter the rate of aneurysm burden.⁴³ Chronic recurrent granulomatous arterial inflammation, with progressive thickening of aneurysm wall in TA is a common finding.²⁹ One study indeed noted wall thickening in most patients (78%) with TA with aneurysm expansion, including all patients with rapid aneurysm progression and subsequent rupture.⁷ Yang et al also noted that onset of vasculitis preceded aortic aneurysm in most patients, with a median delay of 66.0 months,⁶ whereas Pan et al reported the risk of aneurysm to be 4.2-fold higher in patients with a disease duration > 5 years.¹³ This is further supported by persistent histologically active disease found in surgical arterial specimens from patients with clinically and biologically inactive disease.^{20,44}

Our study has several limitations, including its retrospective design and small number of patients. Imaging techniques and follow-up were not standardized, especially in the early years of the clinic. Survival analysis could not be undertaken, as individual relapse dates were not recorded. The latter would have indeed been informative, as length of follow-up varied. Use of antiplatelet drugs, anticoagulants, statins, and/or angiotensin-converting enzyme inhibitors was also not recorded.

In conclusion, a significant proportion of our patients with TA were found to have aneurysmal lesions, most often involving the thoracic aorta. Patients with TA with aneurysms experienced more vasculitis relapses. Considering the risk of rupture, aneurysmal disease should be reported systematically in all studies of TA.

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